

Adverse Drug Reaction Data Analysis

Objective: To analyze a simulated healthcare dataset containing patient symptoms, medication use and timelines, identify possible adverse drug reactions (ADRs), correlate symptoms with medicines, identify higher-risk medicines, and assess possible causes.

1. Dataset Overview

Patient	Age	Medicine	Dose	Symptom / Reaction	Onset
P001	45	Amoxicillin	500 mg	Skin rash	2 days
P002	62	Warfarin	5 mg	Bruising	5 days
P003	34	Ibuprofen	400 mg	Stomach pain	1 day
P004	58	Metformin	500 mg	Nausea	3 days
P005	71	Warfarin	5 mg	Bleeding gums	7 days
P006	29	Amoxicillin	500 mg	Diarrhea	3 days
P007	66	Atorvastatin	20 mg	Muscle pain	14 days
P008	51	Ibuprofen	400 mg	Heartburn	2 days
P009	73	Warfarin	5 mg	Nosebleed	6 days
P010	40	Metformin	500 mg	Nausea	4 days

2. Data Analysis

- **Total patient records:** 10.
- **Medicines represented:** Amoxicillin, Warfarin, Ibuprofen, Metformin and Atorvastatin.
- **Repeated reactions:** Nausea occurred in two patients taking Metformin.
- **Repeated medicine-reaction pattern:** Warfarin was associated with bruising, bleeding gums and nosebleed in this simulated dataset.
- **Time relationship:** Most reported symptoms appeared after medicine exposure, supporting temporal association; however, timing alone does not prove causality.

3. Medicine–Symptom Correlation

Medicine	Observed symptoms	Interpretation
Amoxicillin	Skin rash, diarrhea	Known types of reactions; consider allergy or gastrointestinal intolerance.
Warfarin	Bruising, bleeding gums, nosebleed	Bleeding-related events warrant careful assessment because anticoagulation ca
Ibuprofen	Stomach pain, heartburn	Compatible with gastrointestinal adverse effects.
Metformin	Nausea	Compatible with common gastrointestinal intolerance, especially around treatme
Atorvastatin	Muscle pain	Musculoskeletal symptom that may warrant assessment for statin-associated mu

4. Higher-Risk Medicines Identified

In this simulated dataset, **Warfarin** deserves particular attention because several records involve bleeding-related symptoms. The finding is a signal for further review, not proof that warfarin caused every event. Other medicines should also be assessed using the patient's history, concomitant medicines, laboratory results and clinical information.

5. Possible Causes / Contributing Factors

- Dose and duration of treatment.
- Patient age and underlying disease.
- Concomitant medicines and possible drug–drug interactions.
- Previous allergy or intolerance.
- Renal or hepatic impairment where relevant.
- Disease-related symptoms or other non-drug causes.
- Medication adherence and recent changes in therapy.

6. ADR Assessment Summary

Case	Suspected ADR	Assessment
P001	Amoxicillin → rash	Possible; temporal relationship present, but alternative causes must be excluded.
P002	Warfarin → bruising	Probable in this simulated case because the symptom follows exposure and is pharm
P005	Warfarin → bleeding gums	Probable; requires clinical review and assessment of anticoagulation status.
P007	Atorvastatin → muscle pain	Possible; further history and clinical/laboratory assessment needed.

7. Conclusion

The simulated analysis demonstrates how patient symptoms can be correlated with medication exposure and timelines to identify potential ADR signals. Warfarin-related bleeding symptoms were the most clinically important pattern in this dataset. Final causality should be determined using complete clinical information and a standardized pharmacovigilance method. The dataset is simulated for educational purposes and does not represent real patient records.

Prepared as an educational pharmacovigilance data-analysis report.